

Photometric / Point Transformations

Exam Cheat Sheet — Log • Gamma • Piecewise • Histogram

1. Core Idea: Point Transformation

A point transformation changes each pixel independently using a function. The output pixel depends only on the corresponding input pixel.

General form:

$$s = T(r)$$

- **r** = input/original pixel intensity
- **s** = output/new pixel intensity
- **T** = transformation function

8-bit grayscale: intensity normally ranges from 0 (black) to 255 (white).

2. Log Transformation

Main purpose: Enhance details in dark/low-intensity regions and compress very high intensities.

$$s = c \log(1 + r)$$

The +1 prevents $\log(0)$ when $r = 0$. **c** is a scaling constant.

Key effect: low intensity values are expanded more than high intensity values.

Example: If $s = 50 \log_{10}(1+r)$ and $r = 9$:

$$s = 50 \log_{10}(10) = 50.$$

Remember: Dark image / hidden shadow details → **LOG** transformation.

3. Gamma (Power-Law) Transformation

Main purpose: Control image brightness using a power function.

$$s = c r^\gamma$$

Often r and s are normalized to the range 0–1. The value of γ controls the brightness.

Gamma value	General effect	Shortcut
$\gamma < 1$	Image becomes brighter	Brighten
$\gamma = 1$	No change if $c = 1$	Same
$\gamma > 1$	Image becomes darker	Darken

Example: $r = 0.5$, $c = 1$.

$$\gamma = 0.5 \rightarrow s = (0.5)^{0.5} \approx 0.707 \rightarrow \text{brighter.}$$

$$\gamma = 2 \rightarrow s = (0.5)^2 = 0.25 \rightarrow \text{darker.}$$

Remember: $\gamma < 1 \rightarrow$ brighter; $\gamma > 1 \rightarrow$ darker.

4. Piecewise Transformation

Main idea: Divide the input intensity range into sections and use a different rule/formula for each section.

Common use: contrast stretching and intensity-level slicing.

Contrast Stretching

If useful image intensities are concentrated in a narrow range, we can expand that range to improve contrast.

Example: Map input $50 \rightarrow 0$ and $150 \rightarrow 255$.

$$s = [255 / (150 - 50)](r - 50)$$

For $r = 100$:

$$s = (255/100)(100-50) = 127.5 \approx \mathbf{128}.$$

Intensity-level slicing: Highlight a selected intensity range while suppressing or treating other ranges differently.

Bit-plane slicing: Separate the binary bits of pixel values to study the information contained in individual bit planes.

Remember: Different intensity ranges $\rightarrow$ different rules $\rightarrow$ **PIECEWISE**.

5. Histogram

Histogram: A graph showing how many pixels occur at each intensity value.

X-axis: pixel intensity (e.g., 0–255)

Y-axis: number/frequency of pixels

Image appearance	Typical histogram
Dark image	More pixels toward the left / low intensities
Bright image	More pixels toward the right / high intensities
Low contrast	Pixels concentrated in a limited intensity range

6. Histogram Equalization

Main purpose: Improve contrast by remapping intensity values according to their cumulative distribution.

$$s_k = \text{round}[(L - 1) \times \text{CDF}(r_k)]$$

- **L** = number of possible intensity levels
- **CDF** = cumulative distribution function
- s_k = new/output intensity

Simple example: If $L = 8$ and $\text{CDF} = 0.5$:

$$s = \text{round}[7 \times 0.5] = \text{round}[3.5] \approx \mathbf{4}.$$

Important: Histogram equalization is not the same as contrast stretching. Stretching expands a chosen intensity range; equalization uses the cumulative histogram/CDF to remap intensities.

7. Quick Comparison

Transformation	Main purpose	Formula / clue
Log	Enhance dark-region details	$s = c \log(1+r)$
Gamma	Control brightness	$s = c r^\gamma$
Piecewise	Control specific ranges / contrast	Different rules for ranges
Histogram	Analyze / adjust intensity distribution	Histogram + CDF

8. Exam Memory Tricks

- **LOG** $\rightarrow$ dark details $\rightarrow$ logarithm
- **GAMMA** $\rightarrow$ brightness $\rightarrow \gamma < 1$ bright, $\gamma > 1$ dark

- **PIECEWISE** → different intensity ranges → different rules
- **HISTOGRAM** → distribution of pixel intensities
- **HISTOGRAM EQUALIZATION** → use CDF to improve contrast

9. What to Identify in a Question

Question gives...	Think of...
$\log(1+r)$	Log transformation
r^γ or power γ	Gamma transformation
Different formulas for different ranges	Piecewise transformation
Frequency of intensity values	Histogram
CDF / cumulative probability + L	Histogram equalization

One-line revision: Log handles dark details, Gamma controls brightness, Piecewise handles selected intensity ranges, and Histogram describes/adjusts the distribution of intensities.